

Answers:

1. *Define sine, cosine, and tangent in terms of a right-angled triangle.*
 - a. In a right angled triangle, $\sin(\theta)$ is the length of the opposite side divided by the hypotenuse, $\cos(\theta)$ is adjacent divided by hypotenuse, and $\tan(\theta)$ is opposite / adjacent.
2. *How are the sine and cosine values of an angle related to a point on the unit circle?*
 - a. Generally, you're looking to understand that a point $P(x, y)$ on the unit circle, corresponding to angle θ , has $x = \cos(\theta)$ and $y = \sin(\theta)$.
3. *Explain the relationship between $\sin(\theta)$ and $\cos(\theta)$.*
 - a. They represent sides of the right angled triangle formed by the angle θ . If you were to sketch the graphs of $\sin(\theta)$ and $\cos(\theta)$, you'd see that they make similar graphs, just offset from one another. In fact, $\sin(\theta) = \cos(\theta - \pi/2)$.
4. *Why do we often ignore the hypotenuse when talking about a unit circle?*
 - a. The unit circle has a radius of 1, thus $\cos$ and $\sin$ of angles correspond directly to the x and y coordinates of the point on the circle.
5. *Given a point $(0.707, 0.707)$ on the unit circle for some θ , compute $\sin(\theta)$ and $\cos(\theta)$.*
 - a. $\sin(\theta) = 0.707$, $\cos(\theta) = 0.707$
6. *Given a point $(0.5, 0.866)$ on the unit circle for some θ , compute $\sin(\theta)$ and $\cos(\theta)$.*
 - a. $\sin(\theta) = 0.866$, $\cos(\theta) = 0.5$
7. *Sketch the graph of $\sin(\theta)$ and $\cos(\theta)$ between $\theta = 0$ and $\theta = 2\pi$. How are they related?*
 - a. The important thing here is to notice that these are essentially the same graph, but just shifted. If you look at $\cos(\theta)$, it's the exact same as $\sin(\theta)$ but just shifted over. If you were to shift $\sin(\theta)$ to the left by $\pi/2$, they would overlap exactly.
8. *What is $\cos(0)$, explain how you derived this from the unit circle.*
 - a. So for $\theta = 0$, the corresponding point on the circle is $(1, 0)$, and thus $\cos(\theta) = 1$
9. *Explain why $\tan(\theta)$ approaches infinity as θ approaches $\pi/2$.*
 - a. Recall that $\tan(\theta) = \sin(\theta) / \cos(\theta)$. As θ approaches $\pi/2$, $\cos(\theta)$ approaches 0, and thus $\tan(\theta)$ approaches infinity, since you're dividing by a vanishingly small number.